

ActInf Textbook Group ~ Cohort 5 ~ Session 5

(Chapter 2, Part 2) ~ 10/17/2023

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all right welcome back everyone it is October 17th and we're in our second discussion on chapter two so let's head over there and um first does anybody interesting little colors were added does anyone want to give any opening thoughts or Reflections on chapter two or just bring up some question so I could share briefly comparison to the previous sections uh it took me longer for sure and I had to do more uh search for more reference material because I think it's the first section that's really trying to introduce the the mathematics um of the framework and uh the the first Parts where we're discussing uh very free energy which is only taking into account information about the past and the present to um to compute uh free energy um this was relatively straightforward um I think the section on expected free energy is a more significant leap and I think I need to spend more time I'm um uh analyzing that and and really making sense of of that part of of the chapter awesome thank you anyone else want again just any part of two that stuck with him or a question they have or a a thought on the role of the low rad um I will just uh Echo the sentiment that um I I think I I finally you know um got the handle on free energy as a concept but the expected free energy is still completely impenetrable making that that that um uh cognitive leap is is is is still a little bit beyond beyond my Reach cool we can definitely look over it okay at the at the uh at the very small font overview we have perception as inference so it's still passive is even though we can discuss there are some ways of action is related to even what might be seen as passive inference background on basian statistics example on the background on basian statistics which we can look at again but there's an object in a person's hand and they're updating their beliefs about what they think that object is based upon what they observe it might be an apple it might be a frog and then they see it jump and they update their belief from thinking that it's probably an apple to probably a frog um that case is carried through in the exact basian setting and for smaller State spaces and smaller scale computational problems you can utilize exact base however for larger State spaces you can't always do exact Bay and so that requires an

approximation method and variational free energy or the evidence lower bound is a common approximation method for doing variational bayesian inference. This approximation method is then mobilized in this particular cybernetic setting which leads to a discussion of action. Still we're in the cybernetic setting continuing to discuss like perception and action as part of this unified approximation challenge. Then we get to the real equations. Equation 2.5 and 2.6 are like the two key equations in this chapter. 2.5 is the variational free energy so here is where we have the ability to build on people's work previously and also make massive contributions with our studying here in 2. Five previous cohorts have used the ontology to provide natural language descriptions of the equation so if this just looks like a lot of symbology then it may be helpful to read the ontology description of it. 2.5 variational free energy is about beliefs and data so variational free energy is like the real time sense making Vibe check it's very closely related to how surprising incoming observations are given beliefs. Some more discussion on how the kale diversions comes into play and then it goes from consideration of the real-time variational free energy into the future oriented expected free energy and planning as inference with equation 2.6 but that's not something that all cognitive systems necessarily do. Not all systems engage in explicit planning and then there's some discussions about special cases of expected free energy and that's the end of the low road so chapter 2 pretty much starts with a simple exact Bayes setting slides over to the approximate variational case applies it to cybernetics and then introduces the two key equations in 2.5 and 2.6 variational and expected free energy just to kind of give an overview but but like so as as you have highlighted like those are the exact things to pick up in the low road so anyone else want to just bring up any thought or question or what what area can we go to yeah I think maybe before we move on to the more technical uh details I think it's important to highlight the significance of this jump like it's a conceptual leap that was very striking to me and I think after you spent some time studying this kind of area it's it becomes kind of intuition but but this idea of perception as inference in contrast to what at least throughout my life the way that I've thought about seeing things has been this the straight transaction from from objects in the world to uh just lights on my retina and then some processing processing through my brain and then just kind of this image being projected somewhere um but the the predictive aspect the the inferential top down component of it is is I think it's a remarkable idea and it's interesting that um this isn't it's not really something that you can tell at least I I have couldn't tell just by observing how I look at the world at least not earlier in my life so so I'm I just wanted to

highlight that as a as an interesting uh but significant like a significant leap I think in in thinking about perception yes Sarah and then anyone else hello I have a disclosure a comment and a question so my disclosure is I read the wrong chapter I missed two weeks so I thought I'd missed two chapters but obviously I didn't uh I did read this chapter but I read it like six months ago so I'm just flicking through very fast now so that my disclosure my comment is yeah I totally get the idea it's such a it's such like a a flip to go from this idea of light coming in your eyes I remember in primary school being told off for doing it the wrong way feeling like light goes out and she's like no it goes in and like we we kind of grow up with this idea like we're sensory machines but then I feel like once you make the flip it's all predictive you it's really actually hard to go back and like not see it anymore and then my question was so is I guess it does relate to this chapter which is lucky because you know that's what we're actually on so like figure 2.5 right so You' got this idea of the perception and the action and one thing I've been struggling with is the how they're different right because perception changing your perception is actually an internal action so it's you're updating a model well what's that that's changing a synaptic way that's an action but it's just being performed internally so one thing I've didn't realize I was struggling with this till today is uh yeah how why is perception separate to action and I know like the idea is it's a unified theory of perception action but why do we still have those and how are they different am I missing something so thank you yes yeah okay H how how are perception and action um similar and different and related okay and Kristoff um so I just want to say kind of couple things I like this idea of this what s just said about light light going out rather than and reminded me of a quote from uh kard that life has to be lived forwards but uh can only be understood backwards and there's there's something about the way we explain the world um you know to ourselves or other is that has this kind of like inversion quality to it and and and um I kind of I can't put my finger on it precisely yet but I I see there's a kind of some kind of universal phenomenon that's that's lurking there but um about this leap from you know uh perception to to inference I remember one of the uh earlier papers by Carl Frist he mentioned something really fascinating and he mentions the hubel and wisel um uh experiments with um cat retina I don't know if you um if all of you are familiar with that but they basically were Shining Light um uh onto Cuts uh retina an aized cat and measuring response from some um um some neuron there for you know measuring electric response and they found these recept these ideas receptive fields that responded to various kinds of orientations and and and so on and the interesting phenomenon that they couldn't explain was end stopping so when you have a receptive field you know like it will it will

respond to the specific neuron will respond to some specific orientation and specific length of the um of the rectangle that we're projecting but when the rectangle gets wide enough it stops and the question was why and uh Neuroscience couldn't quite explain this but uh uh sort of you know predictive um the sort of predictive framework um uh I remember Carl mentioned that in one of his papers it starts to suddenly makes sense because kind of in nature you don't really see short lines so when you're projecting a when when you're looking at the Horizon it's always a very very long line but if it becomes shorter then it's you know it's something that um does not occur in nature and therefore for the neuron fires because it actually is a prediction error so it's kind of fascinating that uh um this this inversion of this idea suddenly suddenly made a phenomenon um you know uh explainable which I thought was really really interesting yes so first this uh like coming in and out of the eye um it is really a different view on perception when as as um was brought up like whether we think about well there's this unrolling generative model that's getting light touches from the sensory information getting fine-tuned that's the sort of like inside out view of experience as opposed to a sensory processing I mean sensory processing even as a as a framework makes it sound like the information is out there and we're going to like process the wheat separate the wheat from the chaff and then we're going to end up with like just the real you know we're going to have the pixels on the camera and then we're going to process the pixels and figure out whether it's a cat or a dog or um even further like we're going to process the pixels and then we're going to figure out what to do um as opposed to we're going to have this unfolding model about how things are and their likelihoods and and what to do and then we're going to get get just light touches from sensory data and um I guess the trip is that um if we believe that sensory processing is outside in the world will confirm our beliefs I think there's a few truth drops just in the visual experience that help us understand the Inside Out perception so the first is the blind spot which we don't attend to it's filled in perfectly even though there's a large um patch on the retina that doesn't get that much visual input second one is that in the periphery there are uh fewer um there's lower resolution vision and there's no color vision however in healthy Vision we have color and equal resolution and there's experiments that show like actually well it is lower resolution in the periphery but it's not attended to as such and then we just look over and then the third aspect is the Cade so multiple times per second our eyes are moving and with active inference we can understand okay it's like we're when the Gaze is fixed our Precision is high and we're getting we're paying more attention to visual input and then right before we're ready to execute a cade of course this is subconscious Precision is alleviated the

eye moves and then it's like the the foot goes back on the gaps so the fact that the blind spot the periphery and the saccades are not attended to is is very um tangible that just our visual experience is what we generate it to be with of course information coming from what's external but it's not being processed merely from what's external okay so let's go to s's question here um how are action and perception similar different or related what's anything that that anyone wants to add on this or there's a few pieces to bring in the part about the planning um you know is that in between or is that part of action got great question well so um uh Kristoff brought up the Hubble and whizzle experiments with the cat retina and the selectivities um in different brain regions and that actually connects to one of the most cited uh Neuroscience papers certainly within the ACT INF space which is r and Ballard 99 which provided some of the early strong evidence for predictive coding and so like well so in that retina experiment they had classical patterns like classical cell types oh this one responds to vertical lines this one responds to diagonal lines this one's horizontal or so on however there were these receptive fields like patterns of activation of visual neurons that were extra classical like they were not accounted for within a classical receptive field model and this paper with empirical evidence and and mathematical modeling established that predictive coding which is to say like expectations about visual input were able to account for some of these extra classical phenomena um Andrew oh yeah I just wanted to say just very generally on the the action perception relationship um active inference seems to take the stance that like we are actively inferring information around us and so rather than being like passive observers who are just sitting there we we engage in actions in order to um kind of impact our perception such as like you pick something up and look at it like you can look at it from different angles all of those are like actions that you're taking um that's that's my general sense of the relationship between the two as opposed to some classical view that we're just you know sitting there we're taking in whatever the supposed reality is around us we're not actually doing anything we're just observers um that's it I don't know if that tracks yeah thank you um Jesse then Manuel yeah I think I mean isn't the the two options either you change your mind or you change the world to minimize the free energy so um that's kind of how I see it anything that you do that act that changes the World materially falls in the category of action where um changing your mind is perception I guess what I'm struggling with is that your mind is in the world right so you say you change your mind but that's the same thing as changing the world like I'm still struggling with the the difference because changing your mind it's just okay even even if we reduce it to it's changing firing rates it's changing patterns in in neurons it's not

some esoteric mind concept it's still something physical we're changing there's an action there to change the synchrony between different brain regions or whatever it is that causes that sort of perceptual switch but it's not like an esoteric concept it's still something physical that's changing is what I'm struggling with and but I completely get what you're saying like I do agree with that but then when you get the smaller and smaller you get it's like what is changing your mind it's just changing weights between neurons it's changing synchrony it's changing something physical unless unless you know maybe this will be the answer and we'll find something metaphysical finally in the brain somehow awesome menw well and then yeah just to give my two cents on this point I think is really interesting what s is saying but you know in the formalism uh where the free energy principle comes from my understanding is you know you have to define a dynamical system which has many variables and for this thing to apply there has to be a concept of a mark of blanket which will come later in the book but it's like you have a separation of variables that do not have direct contact with other variables within the system only through this layer of interaction and perception and action are different there in the directionality of the influences so perception only influences the internal state of the subsystem you care about uh but doesn't do the opposite and or or the sorry the internal system doesn't influence the perception States and the opposite with action and when you say yeah like the brain also is taking an action in a physical world and that's true you know Freestone seems to mention pretty often how these Mark of blankets I think of them as you know the mataska the Russian doll many layers of these uh Mark of blankets on top of each other so a cell it's doing its own uh active inference but it's so it's had its own Mark of blanket maybe defined by its own cellular membrane and whatnot but it's part of a circuit and that circuit is also has now a more functional Mark of blanket and it and and they take uh take actions towards other systems and and and you know many many scales of this kind of thing but the I think the more precise difference is this of what variables are changing in your system versus what variables you change of what is not the system all defined by what the mark of blanket is always in the in the sense that there has to be something that is that you care about and some and the rest that you don't care about directly yes very interesting and important point which is a simple or narrow technical difference between action and perception is for a given system of interest that we're just going to focus on the inbound dependencies are perceptual the outbound dependencies are action and then with a different blanket or a different system of Interest a given causal influence might be under understood as well you know it was an action for this for the speaker but then it was a

perception for The Listener so then by taking uh uh by by swapping a perspective um we might give a different AC INF ontology tag to the same causal Edge on a base graph um raw and then Andrew yeah I was thinking a bit about this uh after I got to figure 2.2 in the textbook um because they're the figures they're illustrating this um uh this interface or this this influence that uh an agent has uh on the World to uh generate some kind of uh observations why here and in my mind the difference between perception is in action is at least based on the formalism that we're given here is in the access that we have to the hidden States so out in the world there's there's some some process some generative process as they refer to it and and there's no way to observe though that uh the hidden states of that generative process and the hidden states are the causes of whatever uh observations you have so for example if you see an apple there's something out there in the world that is generating Apple like observations on your retina and then uh inside of your mind you have some also you have a generative model of what that uh process is in out in the world but you actually have some um maybe some influence some more direct influence over that uh over the configuration of the generative model and uh so even if you take some actions internally to update that model it's uh it's different than than like uh some proxy action that you take to change your observations out in the world um that's as far as the distinction has have been clear enough to me but I think this idea of a mark off blanket which will probably Explore More um on in the high road to active inference uh I think that's also a deep an important idea for establishing what it means to be uh a system what it means to be a thing that that is separate in some uh significant way in some statistical way perhaps from from the external world thank you great Point even if we chose to take a kind of active modeling approach to internal action like through attention or whether even conceptualized of learning as action like just the actual parameter updating of learning not just like moving around in the world only through the agent's boundary would the object be intervened upon in the world it'd be like you know if people are watching a movie and they're they might be doing intentional modulation or something they're not intervening with what's on the movie screen versus an outgoing dependency which is to say real over to action then um does modify how hidden States changed their time Andrew sorry um yeah know I think everyone just before me largely Su up what I wanted to say I guess I'll just add um another way I kind of think about it um is like the inclusion of the the idea of preferences and the generative model carries preferences um and in differentiating the um the system of Interest like the model versus the process whose true States we don't have direct access to and so on um like you know for me a human is a

generative model like I have references or homeostatic set points or however you want to view those so I want to keep my Hunger low I would like to get some good sleep I would like to have a well regulated body temperature and um while I would say yes my body my mind and so on are in fact part and parcel with the world I'm in the world I don't think that the world around me has the same preferences I do I don't think it's attempting to uh regulate my body and uh maintain my home ostatic set points I have to engage in in action as well as like attempting to get rather tractable accurate perceptions of the World Around Me In order to act and and uh realize my preferences that are you know all of these homeostatic um set points um active inference seems to paint the process around the model potentially is just like an entropic dissipative set of forces that the agent is trying to act against um in some ways and that's not necessarily a big antagonistic picture between us and the world around us but in some ways yeah we do have to kind of um fight for our lives something like that so yeah great points um you could specify a generative model where the environment was just at the exact temperature that the agent preferred or you could have one where the environment had some regions that were the preferred temperature and some that weren't and then that would lead to the question of well then how would the agent select where to move and that would take us literally exactly to expected free energy where pragmatic value would um upweight or upregulate policy selection that led to observations aligning with preferences but again to to this top um AC action and perception certainly work together that is a kind of way preactive Insight that that there's this um inextricable relationship between perception and action like the um uh even the opening quote of the chapter by William James is connecting thinking and and doing [Music] um however they're not the same we've talked about how for a given system of Interest their directions are different across a blanket um they're they're also modeled as different in the generative model that we specify like this is looking to figure 4.3 but perception is going to be loaded in between observations and hidden States so perception is going to be just from a from what variables you you actually use it's it's mediating the relationship between observations and unobserved latent causes like here's the thermometer reading and then here's the estimate of what temperature it is in the room that's where perception is whereas action is seen as intervening in how hidden States change through time you're not reaching in and changing how the thermometer reading is related to temperature in the abstract you're actually doing something that modifies how temperature in the world changes through time and then the thermometer readings might be different later so it it action and perception are coming in differently to how we specify the generative model uh Susan or or anyone else yeah so here's

how I'm kind of interpreting the the planning um and this kind of brings in you the ch
changing your world changing your mind versus changing your world um so changing the
world if I'm if if I'm trying to change my world then I'm acting on the environment and if I if it
you know if there's um um and bringing in the expected ambiguity and risk that's what's
emerging and if you know if I'm not able to reach the expectations of what's changing in the
world then that's going to bring me back back to changing my mind so that planning is you
know is um is the the inferences that I have about what my power in the world is and I think
that you to for me that's kind of where I bring in the affordances because you the what you
talked about Andrew was was I interpreted that as you someone feels entitled um that the
world's going to provide them and then they act out in the world world as if they're entitled
they may well you know either have to change their mind or they're going to get reinforcement
that they are entitled is that is that cohere yeah like if there's a door and one expects to be on
the other side of the door then it's like you try to open the door okay if it opens then you
change the world and then you proceed on and as you described like then that was like
confirmed or reinforced whereas if it doesn't open I mean what I got to get back changing my
mind yeah you could keep on trying but at some point if it's not openable there's going to have
to be a belief update that that in that moment at the very least that door can't be open otherwise
it's going to be just you know you're gonna um not drink any water and you're going to die
trying to open that door it's like there has to be a stopping point um and and that brings in the
expected ambiguity plus risk yes let's definitely this yes we'll look at 2.5 2.6 and see what um
where these come in um Andrew hi uh multitasking again um just a quick question this act
action perception conversation reminded me I have some questions about attention and
intention do are those Concepts kind of covered or addressed anywhere in active inference or
this textbook attention absolutely in its more uh technical parametric meaning of attention as
precision as well as in its kind of broader psychological uh meaning intention probably more
of a nebulous concept so not to say that you couldn't assign preferences to be intentions for
example the belief desire intention model that's explored in live stream 46 by Ryan Smith at
all where they associate preferences just straightforwardly um W with they talk about the
generative model in terms of the belief desire intention folk psychology model but in this
textbook it's um going to certainly talk about attention and that's a major topic cool yeah I was
just noticing the other day how um with your perceptions there's certain modalities like your
sense Your Skin Sense and your body sense that can be masked um in terms of perception in a
way that like noise cannot from your conscious experience and I thought that might say

something interesting about the active inference model or the the structure in our brains yes cool yeah that this is uh 6.2 here's like different kinds of observations so Sensor Fusion setting and then attention is it allocated across sensory modalities or how is it or where is it distributed across um or within sensory modalities how are those attentions to observations related to like Bel unexperienced beliefs about the world like you're not actually directly experiencing whether there's a jumping frog but you may experience interception auditory visual Jesse could you look at to Andrew's question could you look at policy as intention more or less because you know it's projecting a series of actions that you aren't you know that you'd like to uh have happen but I was thinking that exact same thing so there's a quote on um page 38 I believe um where they're discussing beliefs over policies as intention and this is related to the expected uh free formalism and yeah I think this is an interesting quote to discuss if people have any thoughts about it nice great reference there I don't know if this is two two general the statements but it it to to go from something that is uh Technical and within the ACT INF onology and then kind of like zoom out or adapt it to a more nebulous setting that's that's on you like if you choose to take that interpretation of a given aspect of active inference in a setting then it'll it'll be valid in the ways that it's valid for that setting and it won't be the whole story in the ways that it's not the whole story for that setting however that kind of a question is difficult to answer in the abstract because when we're talking about policy selection we're talking about metronomes and particles of dust and computers and groups of people so so then if but so if someone's thinking well intention I'm meaning like what I experience when I want to do something then you're already in the case of a given system of interest at which point it's no longer like a general question about the ACT INF onology it's more about how you're adapting a term or an equation or something to a given setting at which point it'll be useful how it's useful and it'll be not the whole story in the ways that it isn't um Andrew sorry I never um unraised my hand but I I do actually have a comment on that um related to policy one thing I maybe this is discussed in later uh chapters but policy generation it seems like at least in these early equations there there a set of policies is in inferred but I don't see how policies are generated is is that discussed or implied somewhere yeah good question it connects to the where is planning so policies just as a first pass answer policies are the affordances that are available multiplied by how many uh to to the exponent of how many time steps the planning Horizon is so if their affordances are up down left right there four affordances over one time step there's four policies that can be selected up down left right so for a Time Horizon of one affordances and policies are identical for a Time Horizon of two you'd have 16 policies of

length to up up up down up right up left Etc um so in this setting policies are just affordances over a horizon so then how do I'm not I'm not sure that really answers it unless affordances are generated because it just moves the question absolutely so how are affordances learned or how are they provided so like in the teamas or something like that it's like how does the rat know that it can move up down left right so first we're dealing with maps not territories so just specifying a given Vector of affordances doesn't mean that the rat knows that it can do those affordances or has any kind of like awareness uh about them it's just us describing what we've seen the system do so we're when we're talking about the afford the policy prior or the Habit Vector the prior on um affordances those are something that the modeler infers not necessarily the system of Interest itself however it still is the question of Interest well how are those affordances known and H what happens when it it up down left right and then all of a sudden up down left right and raise up you know an arm or something like that like how do new affordances come into play we can express the question in the ACT infology but write it out for a given system of interest and then I think you'll be well closer to the answer but certainly you can develop nested models where like there's a um a a a context shift that leads to a different set of affordances but but that's a that's um a good question let's look at 2.5 and 2.6 and just look at their partitionings okay so here's 2. five and they write that variational free energy so variational it just means we're using variational methods so we're doing approximate basian computation not exact Baye but approximate Baye using variational methods it may seem an abstract concept but its nature and role it plays in active becomes apparent when decomposed into quantities that are more intuitive and familiar in cognitive science so three lines are different decompositions of variational free energy so they differently partition the kind of same piie in a few ways the top line is more of a statistical physics decomposition that is most like for example Gibbs free energy for those with a chemistry background whereas the second and third line have more to do with Statistics and model fitting and in both cases I mean ultimately in all cases since they're describing the exact same quantity but especially um in this pure statistical second and third decompositions it highlights that we're fitting data but also not one to overfit data and the family of distributions that are being utilized are whatever are used for Q Q is the distribution that we control from the family that we selected so if we say we're going to do variational inference on temperature in this room and we're going to use a gaussian distribution so like a a a bell curve distribution that doesn't mean that the temperature in the room is act changing according to a bell curve or that the thermometer is giving us readings according to that Gan distribution maybe it has like a long tail or maybe it

has some other distribution but it means that we are going to have our hands on the knobs of Q and get to the best fitting which is to say anything wider would be underfit anything tighter would be overfit those are the knobs that we can control it might be grossly inadequate but there is a way to fit the best possible Q to a given Y data and the way to fit the best Q to a y is with reducing the K Divergence shown here and that is variational free minimization if you can reduce that K Divergence between the variational free energy and surprise to zero then variational inference is identical to exact basian inference because minimizing surprise is kind of like the actual Target here if you minimize surprise you've maximized model evidence you know you're getting new thermometer readings coming in if you were minimally surprised by those readings you would have the highest model evidence you would have the best belief so instead of trying to maximize model evidence directly with like a hill climbing approach statistical physics kind of inverts that problem and says well let's think about going downhill with a gradient descent and here is our way to like have a tractable incrementally improvable approach that we know is going to be moving in the right direction to surprise minimization with variational free energy minimization which is highly optimizable again doesn't mean that it's going to get exactly here doesn't even mean it'll get close but you will be able to take incremental improving moves which is kind of shown here so that's the real time beliefs the approximate posterior q and the incoming data y That's the variational free energy and again the first decomposition is more like statistical physics free energies and the second and third decompositions are more like statistics this doesn't have anything to do with the future this is just like a real time unfolding then we get into expected free energy so expected free energy extends active inference to include a prospective form of cognition which is planning so now um rather than just the kind of ball rolling down the hill okay sometimes it jumps left sometimes it jumps right but each time its movement is basically um dictated or constrained by its local surroundings so we could talk about that ball having policy decisions or taking actions in some sort of um very like real- time unfolding way but we might not need to appeal to planning however we may be interested in cognitive models that have an explicit planning component like deliberation or consideration of multiple decisions again this isn't to say that that system does planning just that we're going to make a cognitive model that includes the planning component um key notation is when there's a Tilda little curve curvy line over a variable that is being used to refer to like a Ser a whole time series of that variable over a given time Horizon so now we move from F of Q and Y variational free energy to G of Π so it's different in a bunch of different ways expected free energy looking to the future with these

series of expected hidden States and observations that of course have not happened yet whereas variational free energy was dealing with the actual data why we also noticed that expected free is taking in as an argument policy priors not taking in beliefs and data directly but rather even though we see the Q and the Y but what G is doing most proximately is taking in the prior on policy so this is like a vector of all the possible policies again if it was only one time step that's the same as the affordances if it was two time steps it would be like affordances squared taking in the prior Distribution on those policies and then sharpening or reweighting them according to this so let's think about that up down left right two time step settings so 16 policies so if we had no a priori habit of any of those they would all be like $1/16$ and that Vector would look like 16 tall and each value would be $1/16$ or if we always did policy one and two and then we very rarely did the other ones it would be like 0.5 point 5 and all zeros or you might put that in a computer as like 04949 and then very small values for things that are rarely done G takes in that policy prior and it updates it according to this calculation and so for people who want to look at the code that's very helpful to see where exactly this comes into play takes in that vector and then it sharpens or updates it to give more weight make the policy posterior bigger where where the policy contributes Information Gain or price Matic value pragmatic value Rewards or updates or increases the waiting on policies that bring expected future observations into alignment with preferences that is the definition of pragmatic value the definition of Information Gain or epistemic value is about the difference that you're expected to um get depending on the observations that you expect to see so it's definitely a lot however that is what this top and most common decomposition of expected free energy does and then as Susan brought up another decomposition of G is rather than thinking about what G does as upweighting based upon epistemic imprecise Matic value you could think about G as changing the The Waiting of policies based upon their expected ambiguities and risk um David and then Andrew hi I guess a major a major obstacle for me to understand all of this is literally the mathematic construction of this sentence so like I I for example here I just don't know where the what the ti means in X and Y or what where the C comes from just ask me because I don't know if somebody has this sense of discomfort not being able to decode what what's this trying to tell me and I don't know if somebody have some good resource on how to really understand this this construction yeah thanks for asking totally agreed um hopefully you're in a good resource now and thank you again for asking depending on someone's familiarity with math or computer science and programming there's definitely a lot of approaches but the big approach is many coats of paint like coar grain it just

say Okay 2.6 is expected free energy and you can go down fractal rabbit holes for the rest of your life or at some point you can just say Okay expected free energy is about updating beliefs about action another really helpful resource that again I encourage people to contribute to is the natural language descriptions of the questions it's a lot easier to read there's still a lot of terms in here but the blue terms are ones where we have Mouse over definitions um and then also maybe someone's interested in playing around more with the AI explanations that we used these may be helpful um certainly on your own notebook or however like the notation uh and understanding like you know Q approximate posterior x hidden States Through Time y observations Through Time π policy it is like learning a vocabulary or a symbol matching exercise um and math is a language and acin is a dialect so it just is a lot of exposure and hearing about it Kristoff they pull the equation out of the Hat yeah in in in some ways yes also the textbook is uh is distilling and summarizing across research and there are more lengthy derivations elsewhere the the math learning group again to the extent that people want to do these activities um there's always been like a more fundamentals of math learning group side like just like what does it mean when we have two matrices and then we just like put them next to each other like what what does that mean to multiply them and then also some people like Jonathan and Ali and others have worked on the derivations of the equations and sanj N josi's textbook and other resources help so it's just like but being in the game is the key enabling step um Andrew yeah um actually kind of related to this uh just one aside for some of the Greek characters as soon as I start like memorizing what the pronunciation of the characters are you find that there is a mapping between like me and mind there's some alliteration going on um so that might be a little mnemonic to help um the other thing is I was wondering with the assignment of like um epistemic value to part of these equations is that a generally agreed upon format Within in the sort of scientific body or was that sort of arbitrarily mapped by the researchers um putting these equations together so so like those statistical yeah um beian sort of uh equations is that like agreed upon as um denoting that qualitative uh term in the case of variational free energy yes I'd say it's absolutely non-contentious variational basian methods or energy based methods with these decompositions have been in play for decades expected free energy extends vfe into the discussion of observations that have not happened yet also these are non-contentious decompositions I believe I don't know I don't have any survey data on on who knows or believes what but but I don't think that these are contentious about the latch there is a column here um if you want the raw latch that is a good technique you can paste this into yeah gp4 can use images as input um this is why the stigmar G

is so important the notes that people write in the questions like the more that people add the more material there are for human and non-human Learners um and we make do and we stay in the game with what we have like three or six months ago when we were converting the m to latch this was the best that we could do and it works really well with Cod AI Cod AI does not have the image recognition yet but if slash when they do we'll be able to add another column and just say Okay given what's in the figure dot dot dot dot dot so yeah lot of different interesting emerging ways to work with the material cool so that is um chapter two on the low road for the next two weeks we'll head into chapter 3 on the h so thank you everyone see you next time thank you thank you bye thank you this is great